# Machine Learning Algorithms Laboratory

Experiment 1 Report

| | |
|---|---|
| Experiment No. | 1 |
| Experiment Title | Working with Python Packages - NumPy, Pandas, SciPy, Scikit-learn and Matplotlib [1] |
| Student Name | Gautham S |
| Register Number | 3122247001016 |
| Faculty | Dr. Poreddy Ajay Kumar Reddy |
| Submission Date | |

## 1 Objective

Becoming familiar with commonly used Python libraries used in machine learning. Public datasets were explored and basic Exploratory Data Analysis was performed to understand the data before applying machine learning models.

## 2 Problem Statement

Different datasets were explored using Python libraries such as NumPy, Pandas and Matplotlib. Basic statistics and visualizations were generated to study the data distribution, identify missing values and observe relationships between features. The suitable machine learning task for each dataset was also identified.

## 3 Dataset Description

| **Dataset** | **Description** |
|---|---|
| Dataset Name | Iris Dataset |
| Dataset Source | Kaggle |
| Number of Samples | 150 |
| Number of Features | 4 Numerical Features |
| Number of Classes | 3 (Setosa, Versicolor and Virginica) |
| Missing Values | No Missing Values |
| Purpose | Exploring data using statistical summary and visualizations. |

| **Dataset** | **Description** |
|---|---|
| Dataset Name | Loan Prediction Dataset |
| Dataset Source | Kaggle |
| Number of Samples | 614 |

| | |
|---|---|
| Number of Features | 13 Columns |
| Target Variable | Loan Status |
| Missing Values | Present in a few columns |
| Purpose | Understanding feature distribution and relationships using EDA. |

# 4 Exploratory Data Analysis

## 4.1 Iris Dataset

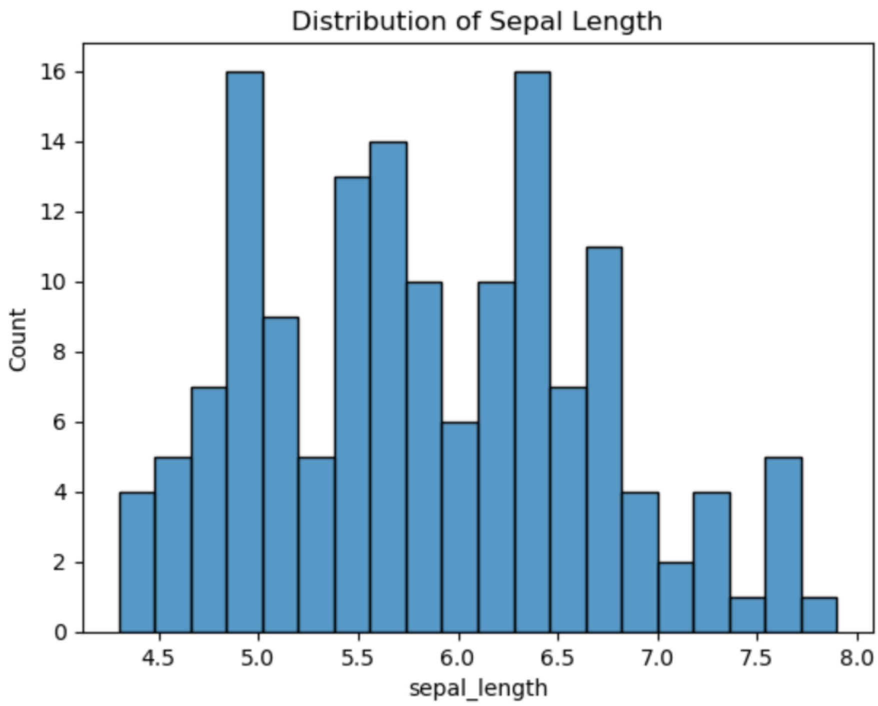


Figure 1: Histogram of Sepal Length

The histogram shows that most sepal length values are concentrated around the middle range. Only a few samples are present at the lower and higher ends.

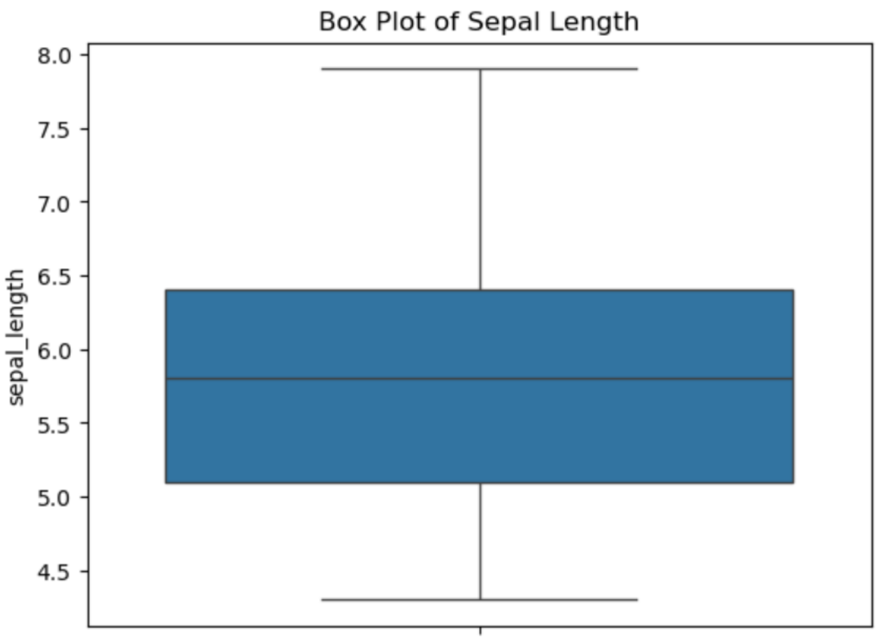


Figure 2: Box Plot of Sepal Length

The box plot shows the spread of the data and indicates a few values outside the normal range. Most observations lie within the interquartile range.

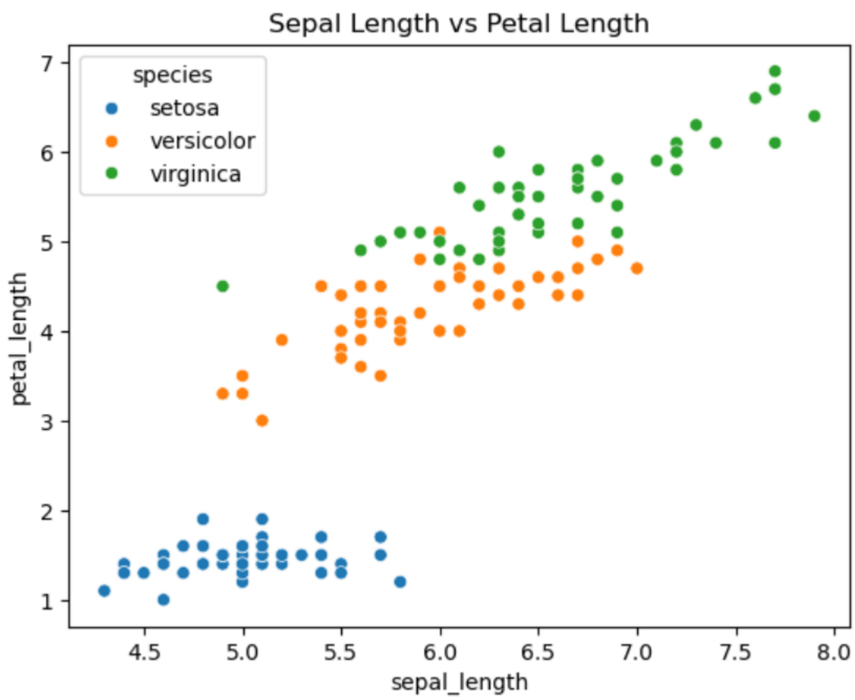


Figure 3: Scatter Plot of Sepal Length vs Petal Length

The scatter plot helps visualize the relationship between two features. Some flower classes form separate groups while a small overlap is also visible.

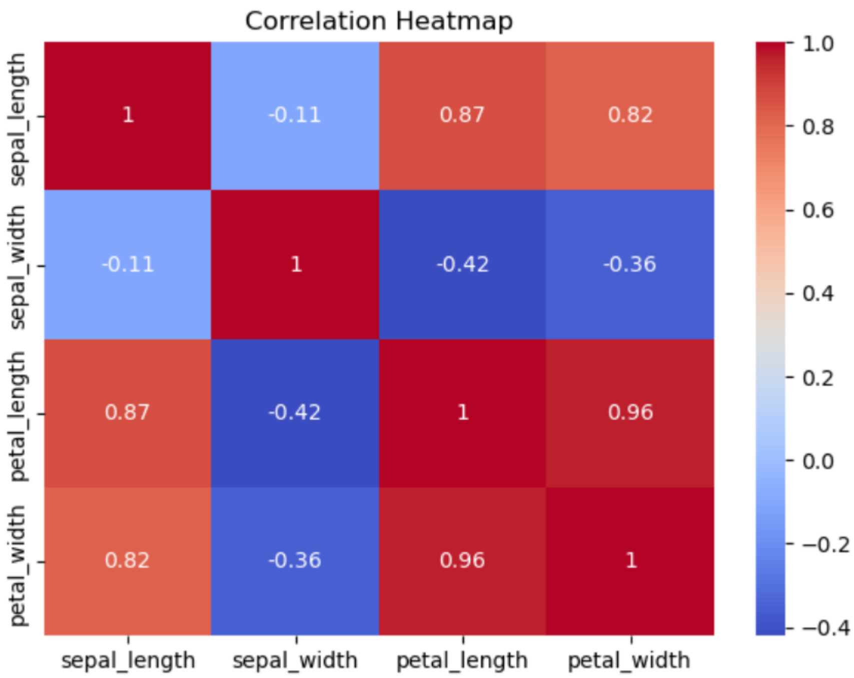


Figure 4: Correlation Heatmap of Iris Dataset

The heatmap shows the correlation between numerical features. It helps identify which features are strongly related to each other.

## 4.2 Loan Prediction Dataset


Figure 5: Loan Status Distribution

The graph shows the number of approved and rejected loan applications. It provides a quick overview of the target variable.

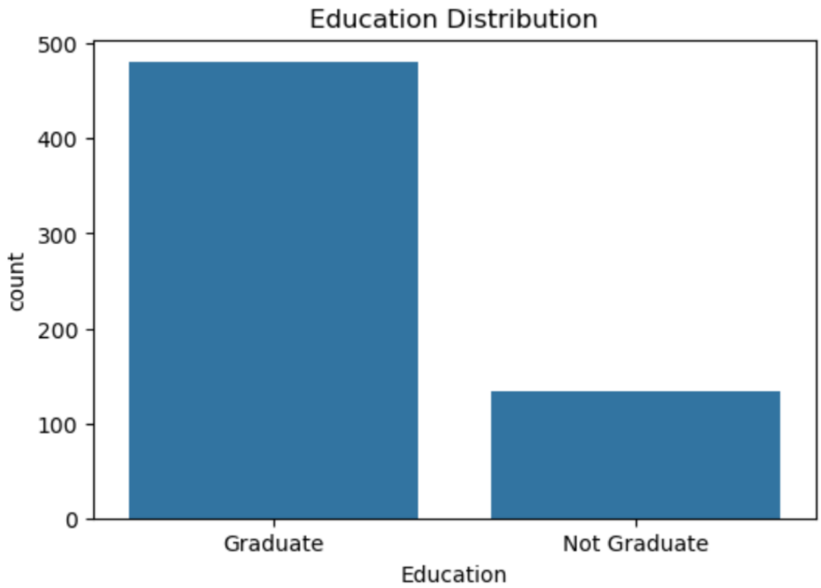


Figure 6: Education Distribution

This graph shows the number of applicants belonging to each education category. It gives an idea of the composition of the dataset.

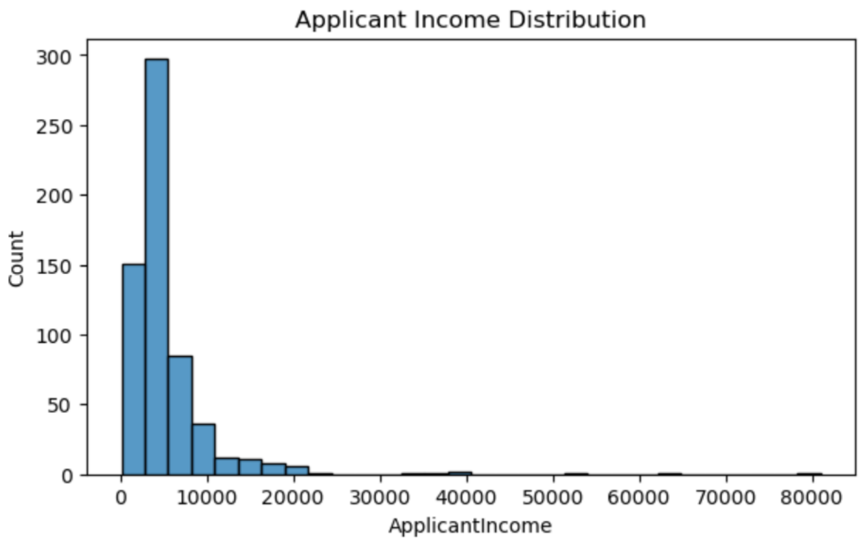


Figure 7: Distribution of Applicant Income

The histogram shows that most applicants have lower incomes, while only a few applicants have very high incomes.

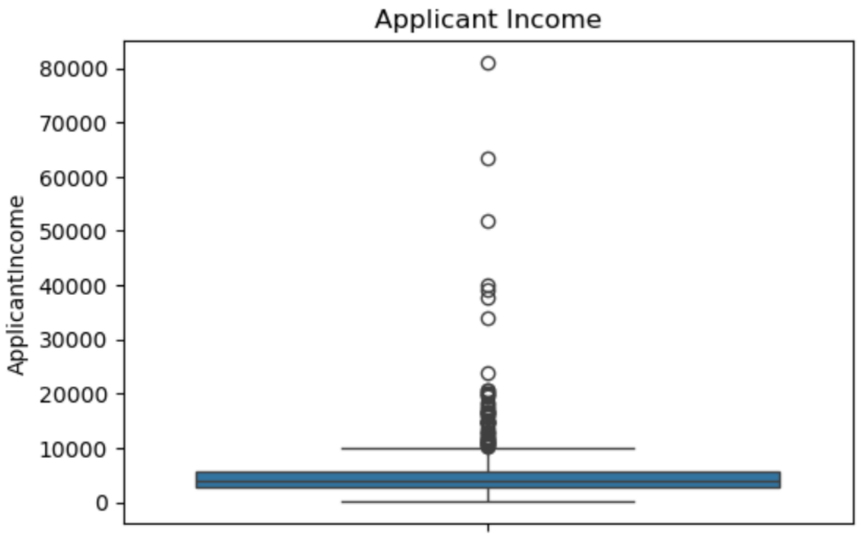


Figure 8: Box Plot of Applicant Income

The box plot highlights the spread of applicant income and shows the presence of a few high-income outliers.

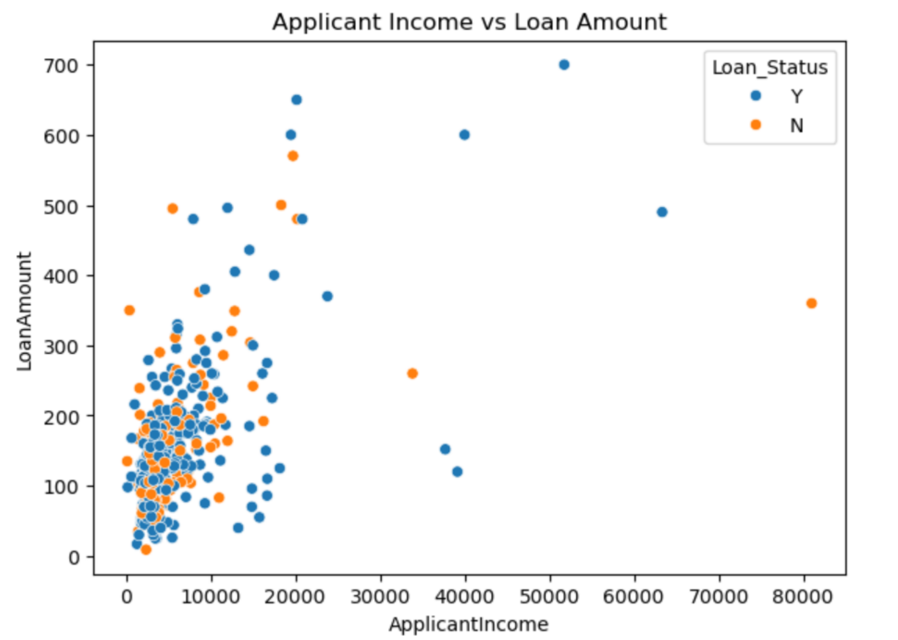


Figure 9: Applicant Income vs Loan Amount

The scatter plot compares applicant income with loan amount. The points are spread across the graph, indicating that loan amount is influenced by multiple factors.

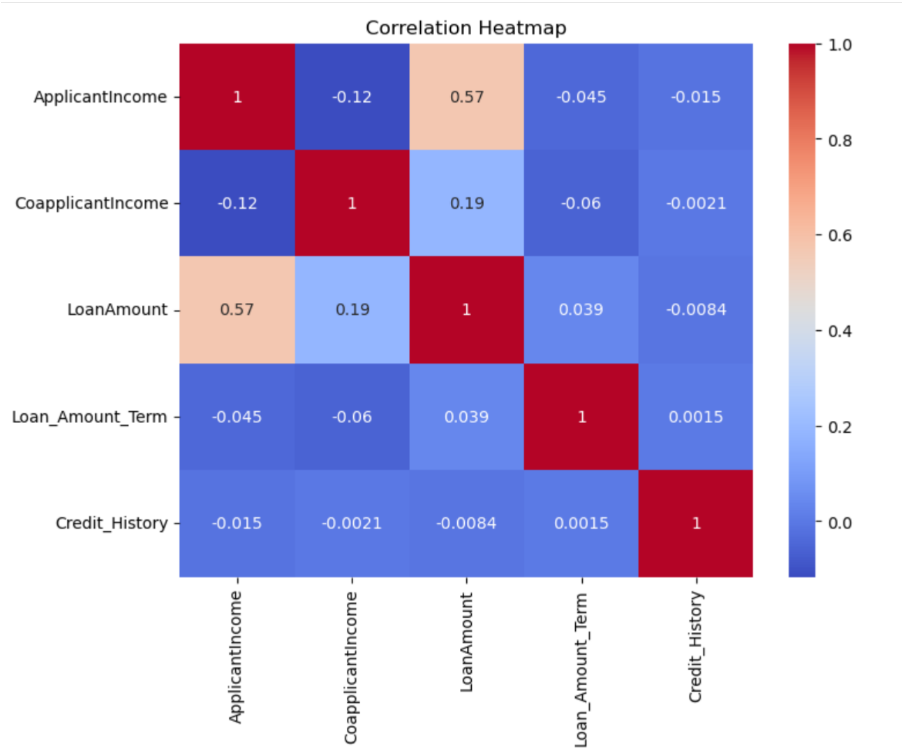


Figure 10: Correlation Heatmap of Loan Dataset

The heatmap shows the relationship between numerical features in the loan dataset. Most correlations are weak to moderate, suggesting that the features contribute different information.

# 5 Data Preprocessing

The datasets were inspected before analysis using Pandas. Missing values were checked for both datasets. The Iris dataset did not contain any missing values, while the Loan Prediction dataset contained missing values in a few columns. Since the aim of this experiment was Exploratory Data Analysis, no additional preprocessing or feature selection was performed.

# 6 Discussion

This experiment provided a basic understanding of working with machine learning datasets using Python libraries. Different plots helped in studying feature distributions, identifying outliers and understanding the relationship between variables. It also highlighted the importance of exploring the data before building any machine learning model.

# 7 Conclusion

In this experiment, the Iris and Loan Prediction datasets were explored using Python libraries such as Pandas and Matplotlib. Exploratory Data Analysis was performed using summary statistics and different visualizations. The experiment helped in understanding the datasets and the initial steps involved in a machine learning workflow.

# 8 References


1. NumPy Documentation. https://numpy.org/doc/
2. Pandas Documentation. https://pandas.pydata.org/docs/
3. Scikit-learn Documentation. https://scikit-learn.org/stable/
4. Kaggle Datasets. https://www.kaggle.com/datasets
5. UCI Machine Learning Repository. https://archive.ics.uci.edu/